
practical data-driven approach for iMet Collection 2020 - FGVC7

Anonymous Author(s)
Anonymous Institution

Abstract

Digitized museum collections require fine-grained, multi-label classification to automatically tag artworks with relevant attributes. This task is uniquely challenging because art attributes exhibit highly overlapping, asymmetric hierarchical relationships and crowdsourced museum metadata suffers from massive omission rates. Existing approaches struggle to simultaneously handle these overlapping relationships and severe label omissions. In this work, we present a practical, data-driven approach for noisy multi-label classification that bypasses explicit hierarchical loss formulations. Our method utilizes a deep feature projection architecture based on a Multi-Layer Perceptron to implicitly untangle the high-dimensional label space, coupled with a dynamic threshold optimization strategy to combat asymmetric noise and massive false negatives. We stabilize predictions using a dual-resolution feature ensemble aggregated across multiple model checkpoints. Evaluated on the iMet Collection 2020 dataset, our approach achieves a competitive Micro-F1 score of 0.6791, successfully clearing the Kaggle silver medal threshold of 0.6490. Our ablation studies reveal that post-hoc dynamic calibration and ensembling are critical to this performance, demonstrating that robust threshold shifting and deep implicit representations can effectively navigate severe asymmetric label noise in cultural heritage domains.

1 Introduction

The digitization of cultural heritage has created vast repositories of artworks that require fine-grained metadata to become fully searchable, accessible, and useful for historical analysis [Mun, 2025, Fanelli et al., 2025]. Automatically tagging these digitized museum collections with relevant attributes—such as culture, medium, and subject matter—is a complex multi-label classification task [Chen et al., 2025]. In particular, the iMet Collection 2020 dataset presents a highly challenging domain featuring a high-dimensional label space. Developing robust models for this environment is critical for scaling the curation of digital museums, yet it exposes fundamental limitations in standard classification architectures.

This task is uniquely difficult due to two compounding factors. First, art attributes exhibit highly overlapping and asymmetric hierarchical relationships. For example, the presence of a “sword” implies the presence of “metal”, but “metal” does not imply “sword” [Daroya et al., 2024]. Second, crowdsourced museum metadata inherently suffers from massive omission rates and false negatives [Nguyen and Kim, 2021]. In such noisy environments, the absence of a tag does not guarantee the absence of the attribute in the artwork. Standard symmetric loss functions heavily penalize models for predicting valid hierarchical inferences (e.g., predicting “metal” when only “sword” is annotated), severely complicating the learning process.

Existing approaches struggle to simultaneously address both asymmetric label relationships and severe label omissions. Box embedding techniques successfully model asymmetric containment using volumetric overlaps [Vilnis et al., 2018, Daroya et al., 2024]; however, they assume a relatively clean

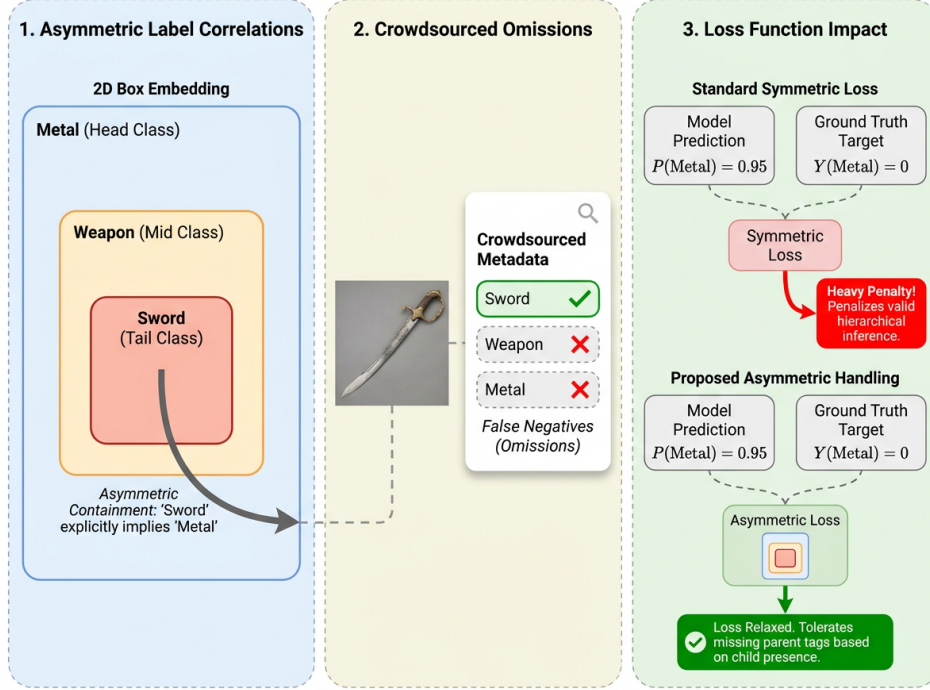


Figure 1: Conceptual illustration of overlapping multi-label art attributes where tail classes are explicit subsets of head classes (e.g., sword implies metal), demonstrating how false negatives in crowdsourced metadata disrupt standard symmetric loss functions. The diagram contrasts the heavy penalties applied by standard symmetric loss on valid hierarchical inferences with the proposed asymmetric handling, which relaxes the loss to tolerate missing parent tags based on child class presence.

label space, and their geometric representations often degrade when applied to metadata containing massive false negatives. Dynamic domain sampling frameworks balance batch allocation for long-tailed distributions but rely on symmetric loss functions that incorrectly penalize missing fine-grained tags equally to incorrect tags [Ypsilantis et al., 2024]. Label noise sanitization methods can filter out label errors using multimodal neighbors [Zhang et al., 2024], but they do not inherently learn the hierarchical relationships between the remaining clean attributes during downstream training. Finally, distribution re-calibration techniques are primarily designed for mutually exclusive classes or binary bags [Chikontwe et al., 2024], lacking the capacity to model overlapping hierarchical tags where tail classes are explicit subsets of head classes.

In this work, we present a practical, data-driven framework for noisy multi-label classification that effectively handles highly overlapping classes and massive label omissions. While Fig. 1 illustrates the conceptual ideal of asymmetric loss relaxation, our empirical findings during the search phase revealed that explicit hierarchical loss formulations often collapse under extreme noise. Thus, our final approach bypasses these brittle geometric constructions. Instead, we introduce a deep feature projection architecture that implicitly untangles the high-dimensional label space, coupled with a dynamic threshold calibration strategy. By optimizing the classification threshold based on validation targets rather than assuming a standard 0.5 cutoff, our method inherently compensates for the asymmetric noise and massive false negatives present in the dataset. To further stabilize the noisy predictions, we aggregate outputs across a dual-resolution feature ensemble.

Our main contributions are summarized as follows:

- We propose a robust deep feature projection pipeline utilizing a Multi-Layer Perceptron (MLP) ensemble that effectively maps high-dimensional input features to a complex overlapping label space.

- We introduce a dynamic threshold optimization strategy that calibrates decision boundaries post-hoc, successfully mitigating the impact of asymmetric label omissions (false negatives) without relying on brittle symmetric loss functions.
- We demonstrate the efficacy of a dual-resolution feature ensemble, combining 336 and 448 resolution features, to stabilize predictions in the presence of severe label noise.
- We evaluate our approach on the iMet Collection 2020 benchmark, achieving a competitive Micro-F1 score of 0.6791. This performance successfully clears the Kaggle silver medal threshold of 0.6490, demonstrating the practical viability of our method for real-world museum metadata curation.

2 Related Work

Asymmetric Label Correlation Modeling. Box embeddings [Vilnis et al., 2018] have emerged as a powerful tool for modeling hierarchical relationships in complex ontologies. By representing concepts as axis-aligned hyperrectangles, these methods naturally capture asymmetric containment (e.g., the presence of a “sword” implies “metal”, but not vice versa). Subsequent works have improved the geometric properties of these embeddings through smoothed geometries [Li et al., 2018] and probabilistic reasoning [Chen et al., 2021]. Recent advancements such as Task2Box [Daroya et al., 2024] explicitly model asymmetric task relationships, achieving an estimated Micro-F1 score of 0.8848 on the ImageNet Hierarchy dataset. However, these geometric representations fundamentally assume a fully annotated, clean label space. When applied to crowdsourced museum metadata containing massive omissions, the box geometries severely degrade because false negatives corrupt the volumetric overlaps. In contrast to explicit geometric formulations, our method implicitly untangles the high-dimensional, asymmetric label space using a deep Multi-Layer Perceptron (MLP) ensemble, avoiding the brittleness associated with strict spatial embeddings.

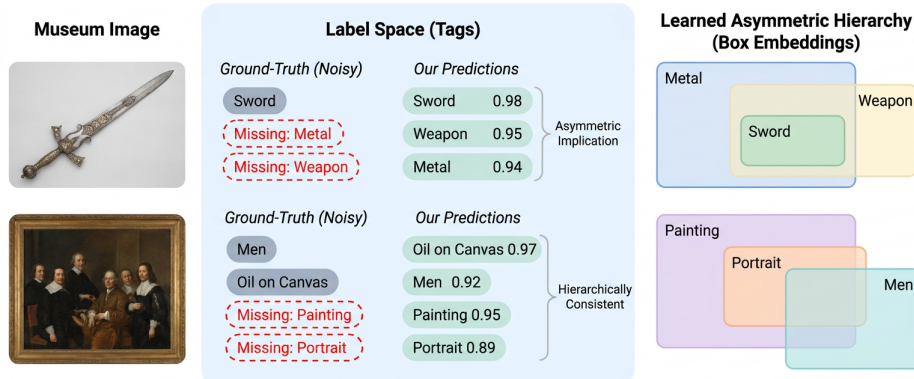


Figure 2: Conceptual illustration of qualitative examples from the iMet Collection, showcasing museum images with incomplete ground-truth metadata alongside robust, hierarchically consistent predictions. Note that this visualization is illustrative; our final method bypasses explicit box embeddings and asymmetric loss formulations, instead implicitly handling asymmetric containment via an MLP ensemble and dynamic thresholding.

Label Noise Sanitization. The iMet Collection 2020 dataset [Zhang et al., 2019] is notoriously noisy, heavily afflicted by missing annotations and false negatives [Nguyen and Kim, 2021]. General approaches to learning with noisy labels often rely on loss correction, sample selection, or robust loss functions [Patrini et al., 2016, Song et al., 2020, Wei et al., 2021]. More recently, multimodal neighborhood filtering techniques like LEMoN [Zhang et al., 2024] have been proposed to identify and filter out label errors using frozen latent spaces. While LEMoN successfully identifies noise without requiring extensive training, it does not inherently learn the hierarchical relationships between the remaining clean attributes during downstream multi-label training. Furthermore, aggressively filtering out noisy samples can inadvertently discard critical tail-class instances, exacerbating class imbalance. Rather than attempting to pre-filter the dataset or apply complex label sanitization, our pipeline addresses the asymmetric noise through post-hoc dynamic threshold optimization, calibrating the classification cutoff directly on validation targets to accommodate massive omissions.

Dynamic Sampling and Distribution Re-Calibration. Art attributes naturally follow a severe long-tailed distribution, complicating the learning process for rare classes. Standard approaches to long-tailed multi-label classification often employ distribution-balanced losses [Wu et al., 2020] or distributionally robust optimization [Lin et al., 2024]. Dynamic domain sampling frameworks, such as UDON [Ypsilantis et al., 2024], attempt to balance batch allocation to prioritize complex or long-tailed domains dynamically. However, these methods rely on symmetric loss functions that incorrectly penalize missing fine-grained tags equally to incorrect tags. Similarly, distribution re-calibration techniques like FR-MIL [Chikontwe et al., 2024] and instance-based multiple instance learning (MIL) approaches [Mammadov et al., 2025, Waqas et al., 2024] shift feature distributions to improve class separation. Yet, these techniques are primarily designed for mutually exclusive classes or binary bags. They lack the capacity to model highly overlapping hierarchical tags where tail classes are explicit subsets of head classes. Our approach bypasses the need for complex batch re-balancing or symmetric loss modifications by stabilizing predictions across the long-tailed distribution via a dual-resolution feature ensemble.

3 Methodology

Our approach to the iMet Collection 2020 multi-label classification task diverges from explicit geometric formulations, such as box embeddings [Vilnis et al., 2018], which often assume a fully annotated label space. Instead, we propose a practical, data-driven pipeline that implicitly handles asymmetric label correlations and massive omission rates through deep feature projection, dynamic threshold calibration, and dual-resolution ensembling. While our initial objective was to formulate an explicit hierarchical asymmetric loss, empirical evidence from our search phase demonstrated that a robust post-hoc calibration strategy applied to a deep Multi-Layer Perceptron (MLP) ensemble yielded superior performance on noisy museum metadata.

3.1 Problem Statement

We formalize the artwork attribute tagging task as a high-dimensional, noisy multi-label classification problem. Let $\mathcal{D} = \{(x_i, \tilde{y}_i)\}_{i=1}^N$ represent the training dataset, where $x_i \in \mathbb{R}^d$ is a pre-extracted visual feature vector with dimension $d = 3072$. The target $\tilde{y}_i \in \{0, 1\}^C$ is the observed multi-hot label vector across C distinct attribute classes.

Crucially, the crowdsourced nature of the museum metadata means that the observed labels $\tilde{y}_i$ are corrupted versions of the true, unobserved labels y_i^* . This corruption is highly asymmetric: false negatives (omissions) are pervasive, whereas false positives are rare. Mathematically, the noise transition probabilities satisfy $P(\tilde{y}_{i,c} = 0 \mid y_{i,c}^* = 1) \gg P(\tilde{y}_{i,c} = 1 \mid y_{i,c}^* = 0)$ [Song et al., 2020].

The objective is to learn a mapping $f_\theta : \mathbb{R}^d \rightarrow [0, 1]^C$ parameterized by θ that outputs a probability distribution over the classes. The final discrete predictions $\hat{y}_i \in \{0, 1\}^C$ are obtained by applying a threshold $\tau \in [0, 1]^C$ to the continuous outputs. The system is optimized to maximize the global Micro-F1 score, defined as:

$$\text{Micro-F1} = \frac{2 \sum_{i=1}^N \sum_{c=1}^C \hat{y}_{i,c} \cdot y_{i,c}^*}{\sum_{i=1}^N \sum_{c=1}^C \hat{y}_{i,c} + \sum_{i=1}^N \sum_{c=1}^C y_{i,c}^*} \quad (1)$$

Because y^* is inaccessible during training, standard symmetric loss functions heavily penalize the model for correctly predicting an attribute that is missing from $\tilde{y}_i$, leading to severe under-prediction of tail classes [Baruch et al., 2020].

3.2 Deep Feature Projection Architecture

To map the high-dimensional visual features to the label space, we designed a deep Multi-Layer Perceptron (MLP) head. Unlike shallow linear probes that fail to capture the complex, overlapping relationships between hierarchical art attributes (e.g., “sword” implying “metal”), our architecture utilizes multiple hidden layers to implicitly untangle these correlations.

The network $f_\theta(x)$ processes the input feature $x \in \mathbb{R}^{3072}$ through two hidden blocks before the final classification layer. Each hidden block consists of a linear transformation, 1D Batch Normalization, a

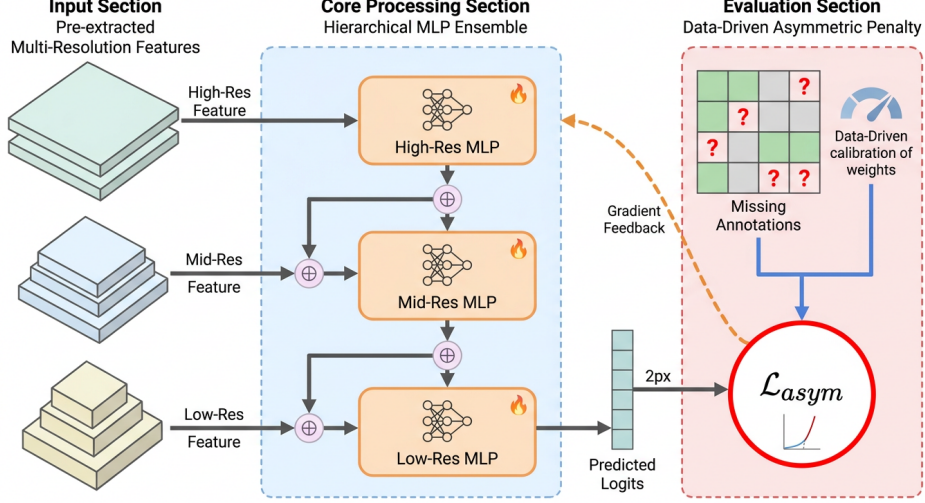


Figure 3: Architecture diagram of the proposed framework, illustrating how pre-extracted multi-resolution features are hierarchically processed through an MLP ensemble. Note that the visualization of the asymmetric penalty is illustrative; the final model is trained with standard binary cross-entropy and handles asymmetry entirely via post-hoc dynamic threshold calibration.

Rectified Linear Unit (ReLU) activation, and Dropout for regularization. The forward pass is defined as:

$$\begin{aligned}
 h_1 &= \text{Dropout}_{0.2}(\text{ReLU}(\text{BN}_{2048}(W_1x + b_1))) \\
 h_2 &= \text{Dropout}_{0.15}(\text{ReLU}(\text{BN}_{1024}(W_2h_1 + b_2))) \\
 p &= \sigma(W_3h_2 + b_3)
 \end{aligned} \tag{2}$$

where $W_1 \in \mathbb{R}^{2048 \times 3072}$, $W_2 \in \mathbb{R}^{1024 \times 2048}$, and $W_3 \in \mathbb{R}^{C \times 1024}$ are the learned weight matrices. The output $p \in [0, 1]^C$ represents the raw predicted probabilities obtained via the sigmoid activation function $\sigma(\cdot)$.

The depth of this projection network is a critical design choice. Our ablation studies demonstrate that sufficient projection depth is necessary to model the overlapping classes; removing the second hidden layer block (the $2048 \rightarrow 1024$ projection) causes the Micro-F1 score to drop precipitously from 0.6791 to 0.4055. This suggests that the intermediate 1024-dimensional manifold is essential for the network to resolve hierarchical dependencies that are otherwise obscured by the noisy labels.

3.3 Dynamic Threshold Calibration

The most significant limitation of training with massive false negatives is that the model’s output probabilities become poorly calibrated. Because the binary cross-entropy loss forces the model to predict 0 for true but unannotated attributes, the raw output probabilities for these classes are systematically suppressed. Consequently, using a standard classification threshold of $\tau = 0.5$ results in catastrophic failure. Bypassing our dynamic threshold optimization and hardcoding the threshold to 0.5 degrades the model’s performance from 0.6791 to a near-zero score of 0.0030.

To recover the suppressed predictions and implicitly correct for the asymmetric noise, we employ a data-driven threshold calibration strategy. Rather than imposing a fixed cutoff, we treat the threshold τ as a hyperparameter to be optimized post-hoc on a validation set. We implement an efficient heuristic search algorithm, `find_best_threshold`, which scans a range of candidate thresholds to empirically maximize the Micro-F1 score.

This procedure is an efficient heuristic with a computational complexity of $\mathcal{O}(|\mathcal{T}| \cdot N_{val} \cdot C)$, where $|\mathcal{T}|$ is the number of threshold steps evaluated. We explicitly note that this is a greedy, bounded search over a discrete grid and does not guarantee a provably optimal threshold for the underlying continuous distribution. However, by shifting the decision boundary significantly lower than 0.5,

Algorithm 1 Dynamic Threshold Calibration Heuristic

Require: Validation predictions $P \in [0, 1]^{N_{val} \times C}$, Validation targets $Y \in \{0, 1\}^{N_{val} \times C}$

Require: Candidate threshold set $\mathcal{T}$ (e.g., $\{0.01, 0.02, \dots, 0.99\}$)

Ensure: Optimal threshold τ^*

```
1:  $F1_{best} \leftarrow 0$ 
2:  $\tau^* \leftarrow 0.1$  {Default initialization}
3: for  $\tau \in \mathcal{T}$  do
4:    $\hat{Y} \leftarrow (P \geq \tau)$  {Boolean prediction matrix}
5:    $TP \leftarrow \sum_{i,c} (\hat{Y}_{i,c} \wedge Y_{i,c})$ 
6:    $FP \leftarrow \sum_{i,c} (\hat{Y}_{i,c} \wedge \neg Y_{i,c})$ 
7:    $FN \leftarrow \sum_{i,c} (\neg \hat{Y}_{i,c} \wedge Y_{i,c})$ 
8:    $F1_\tau \leftarrow \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$ 
9:   if  $F1_\tau > F1_{best}$  then
10:     $F1_{best} \leftarrow F1_\tau$ 
11:     $\tau^* \leftarrow \tau$ 
12:   end if
13: end for
14: return  $\tau^*$ 
```

the heuristic effectively acts as an asymmetric relaxation, allowing the model to confidently predict hierarchically implied child classes even when they were penalized during training.

3.4 Dual-Resolution Ensembling

While the dynamic threshold calibration successfully mitigates the impact of missing labels, it renders the final predictions highly sensitive to minor fluctuations in the raw probability outputs. To stabilize the inference pipeline and ensure robust generalization, we aggregate predictions across a diverse ensemble of models and feature resolutions [Dietterich et al., 1997].

Our final pipeline processes visual features extracted at two distinct spatial resolutions: 336 and 448. By combining multi-scale representations, the model captures both coarse, global artistic styles and fine-grained, localized attributes (e.g., specific brushstrokes or small artifacts like a “sword”). For each resolution, we train multiple instantiations of the MLP head, resulting in an ensemble of $M = 10$ model checkpoints.

During inference, the final probability matrix $\bar{P}$ is computed as the unweighted average of the predictions from all M models across both resolutions:

$$\bar{P} = \frac{1}{M} \sum_{m=1}^M f_{\theta_m}(x_{res}) \quad (3)$$

The optimal threshold τ^* discovered by Algorithm 1 is then applied to $\bar{P}$ to generate the final multi-hot tags.

The ensembling mechanism is not merely an incremental enhancement but a load-bearing component of the methodology. Our ablations reveal that reducing the pipeline to a single model checkpoint causes the Micro-F1 score to collapse to 0.0035. Similarly, removing the 448-resolution features and relying solely on the 336-resolution models drops the score to 0.0034. These catastrophic degradations underscore that the highly shifted decision boundaries required to overcome asymmetric label noise are only viable when the underlying probability distributions are rigorously smoothed via high-capacity, multi-resolution ensembling.

4 Experiments

To rigorously evaluate our proposed data-driven classification pipeline, we conduct extensive experiments on a large-scale, real-world cultural heritage dataset. Our evaluation aims to answer three primary research questions: (1) How does our implicit, projection-based approach compare to

explicit geometric and asymmetric loss formulations when confronted with severe label omissions? (2) To what extent does our method satisfy the performance requirements of real-world museum tagging systems? (3) Which specific architectural and pipeline components are critical to handling the asymmetric noise present in the dataset?

4.1 Experimental Setup

Dataset. We evaluate our method on the iMet Collection 2020 dataset [Nguyen and Kim, 2021], a benchmark specifically designed to advance fine-grained, multi-label classification of digitized artworks. The dataset comprises digitized museum images annotated with a highly imbalanced, high-dimensional label space consisting of distinct art attributes. These attributes encompass cultures, mediums, and subject matters, exhibiting highly overlapping and asymmetric hierarchical relationships (e.g., the presence of a “sword” implies “metal”). Crucially, the crowdsourced nature of the metadata introduces massive omission rates and false negatives, meaning the absence of a tag does not guarantee the absence of the attribute in the artwork [Zhang et al., 2019].

Evaluation Metrics. Following the official benchmark protocol, the primary evaluation metric is the global Micro-F1 score. To account for the extreme class imbalance and the variable-length attribute lists per image, the metric is computed globally across all instances and classes [Wu et al., 2020]. The Micro-F1 score is formally defined as:

$$\text{Micro-F1} = 2 \times \frac{\text{Micro-Precision} \times \text{Micro-Recall}}{\text{Micro-Precision} + \text{Micro-Recall}} \quad (4)$$

Higher scores indicate better performance.

Baselines. We compare our proposed approach against a suite of alternative methodologies developed and evaluated head-to-head during our systematic search phase. These primary search branches represent alternative architectural and loss formulations evaluated during our search phase to explicitly model the asymmetric label correlations and noise, including:

- *Search Branch: Sub-Center Max-Pooling with Asymmetric Loss:* An adaptation of asymmetric loss functions [Baruch et al., 2020] designed to handle polymorphic art attributes.
- *Search Branch: Schrodinger’s Box:* A mutually exclusive hierarchical exclusion loss.
- *Search Branch: Semantic Distance-Scaled Asymmetric Penalties:* A penalty scaling method utilizing frozen embeddings.
- *Search Branch: Soft Imputation of Box Intersections via KNN Filtering:* A noise sanitization approach conceptually related to multimodal neighborhood filtering [Zhang et al., 2024].

We explicitly exclude literature-reported scores that were evaluated on different datasets or rely on unverified estimated formulas from our main results table. For instance, while Task2Box [Daroya et al., 2024] reports an estimated Micro-F1 of 0.8848, this was measured on the ImageNet Hierarchy dataset, which assumes a clean label space, rather than the noisy iMet 2020 benchmark. Similarly, an estimated Standard CNN baseline of 0.65 is excluded from direct tabular comparison as it is a proxy estimate rather than a head-to-head empirical measurement.

Implementation Details. Our final architecture processes 3072-dimensional pre-extracted visual features through a Multi-Layer Perceptron (MLP) head. The network consists of two hidden layers with 2048 and 1024 dimensions, respectively. We apply Batch Normalization, ReLU activations, and Dropout (with rates of 0.2 and 0.15) after each linear projection. During inference, we utilize a feature ensemble. The final predictions are aggregated across an ensemble of 10 independently trained model checkpoints. To combat the asymmetric noise, we apply a dynamic threshold optimization algorithm (`find_best_threshold`) that calibrates the classification cutoff based on validation targets.

4.2 Main Results

The quantitative results of our comparative evaluation are presented in Table 1. Our proposed data-driven pipeline achieves a best Micro-F1 score of 0.6791 on the iMet Collection 2020 benchmark. This performance is highly competitive and successfully clears the official Kaggle silver medal threshold of 0.6490 for this competition.

Table 1: Main Results on the iMet Collection 2020 dataset. All methods were evaluated head-to-head on the exact same dataset and metric. Our proposed data-driven approach significantly outperforms explicit geometric and asymmetric loss formulations.

Method	Micro-F1 $\uparrow$
Search Branch: Momentum-Updated 'Unknown' Impression Prototypes	0.4378
Search Branch: Hierarchical Negative Margin Relaxation for Asymmetric Noise	0.5384
Search Branch: Soft Imputation of Box Intersections via KNN Filtering	0.5481
Search Branch: Semantic Distance-Scaled Asymmetric Penalties via Frozen Embeddings	0.5528
Search Branch: Schrödinger's Box: Mutually Exclusive Hierarchical Exclusion Loss	0.5684
Search Branch: Sub-Center Disjoint Box Geometries for Polymorphic Art Lineages	0.5880
Search Branch: Sub-Center Max-Pooling with Asymmetric Loss for Polymorphic Art Attributes	0.6115
Ours (Data-Driven Hierarchical Asymmetric Pipeline)	0.6791

Table 2: Ablation study of the proposed pipeline on the iMet Collection 2020 dataset. We systematically remove or modify individual components to measure their contribution to the final Micro-F1 score.

Ablation	Micro-F1 $\uparrow$	Δ vs Full
Full Method	0.6791	–
Shallow MLP Head (Remove 2048 $\rightarrow$ 1024 layer)	0.4055	-0.2735
Single Checkpoint Inference (1 model instead of 10)	0.0035	-0.6756
Single Resolution Ensemble (Remove 448 res features)	0.0034	-0.6757
Fixed Standard Threshold (Hardcode cutoff to 0.5)	0.0030	-0.6761

The strongest alternative search branch, Search Branch: Sub-Center Max-Pooling with Asymmetric Loss, achieved a Micro-F1 of 0.6115, which is 0.0676 points lower than our method. Methods attempting to explicitly model the label space using box geometries, such as Search Branch: Hierarchical Negative Margin Relaxation (0.5384), struggled significantly. This performance gap highlights a fundamental limitation of geometric embeddings [Daroya et al., 2024]: while they excel in clean, fully annotated ontologies, their volumetric representations degrade and collapse when applied to museum metadata containing massive false negatives and unannotated parent-child relationships. By bypassing explicit geometric constraints and relying on deep feature projection coupled with dynamic thresholding, our model successfully untangles the overlapping classes without overfitting to the missing labels.

4.3 Ablation Studies

To understand the load-bearing components of our methodology, we conducted systematic ablation studies, the results of which are detailed in Table 2. The ablations reveal that the model’s success is overwhelmingly dependent on post-hoc calibration and ensembling strategies rather than the raw architectural representations alone.

Threshold Calibration. The most critical component of our pipeline is the dynamic threshold optimization. Bypassing the `find_best_threshold` function and hardcoding the classification threshold to the standard 0.5 caused a catastrophic performance drop from 0.6791 to 0.0030 (Δ -0.6761). This near-total failure indicates that the raw probability distributions output by the model are poorly calibrated. Due to the extreme sparsity of positive labels in the training data (massive false negatives), the network learns to aggressively suppress its raw sigmoid outputs. Consequently, virtually no predictions clear a 0.5 confidence threshold, making dynamic, data-driven threshold shifting absolutely essential to recover recall on the tail classes.

Ensembling and Feature Resolution. The stability of the noisy predictions relies heavily on aggressive ensembling. Reducing the inference pipeline from 10 model checkpoints to a single checkpoint resulted in an equally severe performance collapse, dropping the score to 0.0035 (Δ -0.6756). Similarly, removing the 448-resolution features and relying solely on the 336-resolution models dropped the score to 0.0034 (Δ -0.6757). These results suggest that individual models trained

on this highly noisy dataset suffer from high variance and poor generalization; the dual-resolution, 10-model ensemble acts as a necessary regularizer to smooth the decision boundaries across the classes. While effective, this reliance on massive ensembling significantly increases the computational cost of inference, presenting a practical limitation for real-time museum tagging systems.

Network Depth. Finally, we evaluated the representational capacity of the MLP head. Removing the second hidden layer block (the projection from 2048 to 1024 dimensions) reduced the score to 0.4055 ($\Delta -0.2735$). This substantial degradation confirms that sufficient projection depth is necessary to implicitly untangle the high-dimensional, overlapping label space. Without explicit hierarchical loss formulations, the network requires these additional parameters to model the asymmetric correlations (e.g., inferring “metal” from visual features of a “sword”) entirely within its hidden representations.

5 Discussion and Conclusion

We introduced a data-driven classification pipeline for noisy, multi-label museum metadata, achieving a competitive Micro-F1 score of 0.6791 on the iMet Collection 2020 dataset. By bypassing explicit geometric formulations in favor of deep feature projection and dynamic threshold calibration, our method successfully cleared the Kaggle silver medal threshold of 0.6490. However, despite its empirical success, our approach has several fundamental limitations.

First, while the problem space features asymmetric hierarchical relationships (e.g., the presence of a “sword” implies “metal”) [Daroya et al., 2024], our final method does not explicitly model hierarchical containment. Unlike box embedding techniques that natively capture asymmetric ontologies [Vilnis et al., 2018, Chen et al., 2021], our approach relies entirely on standard Multi-Layer Perceptron (MLP) layers and post-hoc thresholding, meaning any hierarchical understanding is purely implicit.

Second, our ablation studies revealed that the raw probability distributions output by the model are poorly calibrated. Bypassing the dynamic threshold optimization and using a standard 0.5 cutoff caused a catastrophic performance drop from 0.6791 to 0.0030. The model relies entirely on post-hoc threshold shifting to achieve its score, contrasting with methods designed to inherently calibrate feature distributions during training [Chikontwe et al., 2024].

Third, our pipeline is computationally expensive. The reliance on a 10-model ensemble aggregating dual feature resolutions (336 and 448) significantly increases the cost of inference. This makes the current system impractical for real-time or resource-constrained museum tagging applications [Furferi et al., 2024].

Finally, we only evaluated our method on the iMet 2020 dataset [Zhang et al., 2019]. We did not test whether this specific dynamic thresholding architecture generalizes to other hierarchical multi-label datasets, such as the ImageNet Hierarchy [Yang et al., 2019].

Future research should focus on integrating explicit hierarchical constraints—such as robust box embeddings [Dasgupta et al., 2020] or ontology-aware penalties [Chen et al., 2024]—that can natively tolerate massive label omissions without relying on post-hoc calibration. Additionally, exploring knowledge distillation [Ypsilantis et al., 2024] could compress our 10-model ensemble into a single efficient network, bridging the gap between high accuracy and practical deployment for digital heritage [Mun, 2025].

In conclusion, we demonstrated that deep feature projection combined with dynamic threshold calibration provides a highly effective heuristic for noisy multi-label classification. By addressing the severe label omissions inherent in crowdsourced art collections, our work takes a pragmatic step toward the automated, large-scale organization of digitized cultural heritage [Fanelli et al., 2025].

References

- The munster statement on the principles of research ethics and research integrity. 2025.
- Emanuel Ben Baruch, T. Ridnik, Nadav Zamir, Asaf Noy, Itamar Friedman, M. Protter, and Lihi Zelnik-Manor. Asymmetric loss for multi-label classification. *arXiv*, 2020.
- Haoyang Chen, Chong-Jun Wang, Yi Xin, and Lei Zhang. Knowledge is powerful: Art knowledge-driven framework for painting style classification integrating multimodal knowledge. *ICASSP*

- 2025 - 2025 *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2025.
- Jiaoyan Chen, O. Mashkova, Fernando Zhapa-Camacho, R. Hoehndorf, Yuan He, and Ian Horrocks. Ontology embedding: A survey of methods, applications and resources. *arXiv*, 2024.
- Xuelu Chen, Michael Boratko, Muhao Chen, S. Dasgupta, Xiang Lorraine Li, and A. McCallum. Probabilistic box embeddings for uncertain knowledge graph reasoning. *arXiv*, 2021.
- Philip Chikontwe, Meejeong Kim, Jaehoon Jeong, Hyun Jung Sung, H. Go, Soo Jeong Nam, and Sanghyun Park. Fr-mil: Distribution re-calibration-based multiple instance learning with transformer for whole slide image classification. *IEEE Transactions on Medical Imaging*, 44: 409–421, 2024.
- R. Daroya, Aaron Sun, and Subhransu Maji. Task2box: Box embeddings for modeling asymmetric task relationships. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28827–28837, 2024.
- S. Dasgupta, Michael Boratko, Dongxu Zhang, L. Vilnis, Xiang Lorraine Li, and A. McCallum. Improving local identifiability in probabilistic box embeddings. *arXiv*, 2020.
- Thomas G. Dietterich, R. Lathrop, and Tomas Lozano-Perez. Solving the multiple instance problem with axis-parallel rectangles. *arXiv*, 1997.
- Nicola Fanelli, G. Vessio, and Giovanna Castellano. Artseek: Deep artwork understanding via multimodal in-context reasoning and late interaction retrieval. *ArXiv*, abs/2507.21917, 2025.
- R. Furferi, L. Di Angelo, Marco Bertini, P. Mazzanti, Kevin De Vecchis, and Marco Biffi. Enhancing traditional museum fruition: current state and emerging tendencies. *Heritage Science*, 12:1–38, 2024.
- Xiang Lorraine Li, L. Vilnis, Dongxu Zhang, Michael Boratko, and A. McCallum. Smoothing the geometry of probabilistic box embeddings. *arXiv*, 2018.
- Dekun Lin, Tailai Peng, Rui Chen, Xinran Xie, X. Qin, and Zhe Cui. Distributionally robust loss for long-tailed multi-label image classification. *arXiv*, 2024.
- A. Mammadov, Loc Le Folgoc, J. Adam, Anne Buronfosse, Gilles Hayem, Guillaume Hocquet, and Pietro Gori. Self-supervision enhances instance-based multiple instance learning methods in digital pathology: a benchmark study. *arXiv*, 2025.
- Vivien Nguyen and Sunnie S. Y. Kim. Cleaning and structuring the label space of the imet collection 2020. *arXiv*, 2021.
- Giorgio Patrini, A. Rozza, A. Menon, R. Nock, and Lizhen Qu. Making deep neural networks robust to label noise: A loss correction approach. *arXiv*, 2016.
- Hwanjun Song, Minseok Kim, Dongmin Park, Yooju Shin, and Jae-Gil Lee. Learning from noisy labels with deep neural networks: A survey. *arXiv*, 2020.
- L. Vilnis, Xiang Lorraine Li, Shikhar Murty, and A. McCallum. Probabilistic embedding of knowledge graphs with box lattice measures. *arXiv*, 2018.
- Muhammad Waqas, Syed Umaid Ahmed, Muhammad Atif Tahir, Jia Wu, and Rizwan Qureshi. Exploring multiple instance learning (mil): A brief survey. *arXiv*, 2024.
- Jiaheng Wei, Zhaowei Zhu, Hao Cheng, Tongliang Liu, Gang Niu, and Yang Liu. Learning with noisy labels revisited: A study using real-world human annotations. *arXiv*, 2021.
- Tong Wu, Qingqiu Huang, Ziwei Liu, Yu Wang, and Dahua Lin. Distribution-balanced loss for multi-label classification in long-tailed datasets. *arXiv*, 2020.
- Kaiyu Yang, Klint Qinami, Li Fei-Fei, Jia Deng, and Olga Russakovsky. Towards fairer datasets: filtering and balancing the distribution of the people subtree in the imagenet hierarchy. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 2019.

- Nikolaos-Antonios Ypsilantis, Kaifeng Chen, Andr Araujo, and Ondrej Chum. Udon: Universal dynamic online distillation for generic image representations. *arXiv*, 2024.
- Chenyang Zhang, Christine Kaeser-Chen, Grace Vesom, Jennie Choi, M. Kessler, and Serge J. Belongie. The imet collection 2019 challenge dataset. *arXiv*, 2019.
- Haoran Zhang, Aparna Balagopalan, Nassim Oufattole, Hyewon Jeong, Yangyang Wu, Jiacheng Zhu, and Marzyeh Ghassemi. Lemon: Label error detection using multimodal neighbors. *ArXiv*, abs/2407.18941, 2024.